User Manual

cAMP Hunter™ Follitropin alfa Bioassay Kit

For Measurement of Follitropin alfa-mediated cAMP Accumulation

2-Plate Kit: 95-0119Y2-00103

10-Plate Kit: 95-0119Y2-00104

Document Number 70-392 Rev1

This page has been left intentionally blank.

Table of Contents

Important: Please read this entire user manual before proceeding with the assay.

Table of Figures

1 Overview

cAMP Hunter Follitropin alfa Bioassay Kits provide a functional, robust, highly sensitive, and easy-to-use cell-based assay to study potency and neutralizing antibodies for Follitropin alfa and its analogues. The bioassay kits contain all the reagents needed for a complete assay including cells, detection reagents, cell plating reagent, positive control agonist, and assay plates. The ready-to-use cryopreserved cells have been manufactured to ensure assay reproducibility and faster implementation from characterization to lot release. This bioassay has been optimized for a 96-well format but can be adapted to a 384-well format if needed.

1.1 Assay Principle

Ligand-mediated GPCR stimulation leads to the activation of G-proteins, which in turn triggers downstream signaling pathways by recruiting, activating or inhibiting cellular enzymes. One such enzyme is adenylate cyclase, which catalyzes the conversion of adenosine triphosphate (ATP) to cAMP. Adenylate cyclase is either stimulated or inhibited by the G-protein subunits, Gαs and Gαi, respectively. In the cAMP Hunter Follitropin alfa Bioassay Kit, cells overexpressing FSHR utilize the natural coupling status of the GPCR to monitor activation of the $G\alpha_s$-coupled receptor. Following stimulation, the functional activation status of FSHR is monitored by measuring the cellular cAMP levels using a homogeneous, gain-of-signal competitive immunoassay based on Enzyme Fragment Complementation (EFC) technology.

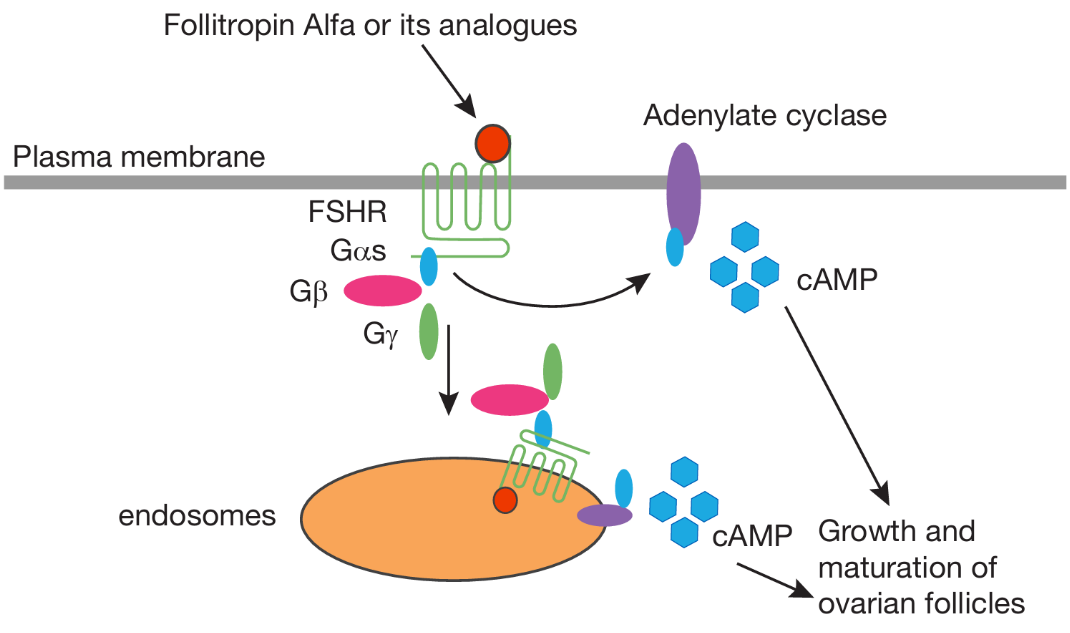
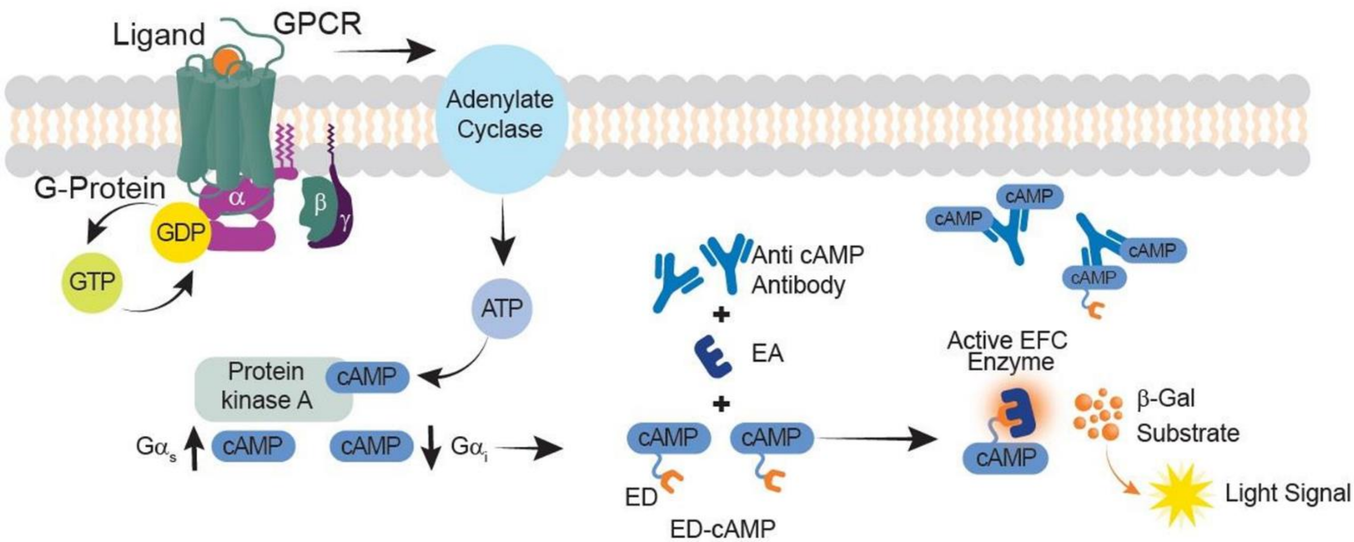


Figure 1: Assay Principle

The cAMP Hunter Bioassay has been developed to interrogate the GPCR cAMP Pathway. Ligand-mediated activation of a GPCR either stimulates or inhibits adenylate cyclase to modulate cellular cAMP levels. In the case of FSHR, its activation by Follitropin alfa stimulates adenylate cyclase, which in turn enables the production of cAMP.

1.2 cAMP Detection Kit Principle

The EFC technology uses a β-galactosidase (β-gal) enzyme split into two fragments, the Enzyme Donor (ED) and Enzyme Acceptor (EA). Independently these fragments have no β-gal activity; however, in solution they rapidly complement to form an active β-gal enzyme. In this assay, cAMP from cell lysates and ED-labeled cAMP (ED-cAMP) compete for anti-cAMP antibody (Ab). Antibody-bound ED-cAMP will not be able to complement with EA, but unbound ED-cAMP is free to complement EA to form an active enzyme, which subsequently produces a luminescent signal. The signal from the assay is directly proportional to the amount of cellular cAMP in the well, i.e., the greater the amount of GPCR activation, the higher the cAMP levels inside the cells, and the larger the signal in this assay.



'Created with BioRender.com'

Figure 2: cAMP Detection Kit Principle

When cellular cAMP levels are low, ED-labeled cAMP successfully binds with the anti-cAMP-antibody, as there aren't enough cellular cAMP molecules to compete against it. The bound ED-labeled cAMP will not be available to undergo complementation. In contrast, when cellular cAMP levels are high, the cellular cAMP molecules bind to the anti-cAMP antibody instead, leaving ED-labeled cAMP relatively free. Upon the addition of the detection reagent containing EA, the ED and EA fragments undergo complementation, which successfully forms an active β-galactosidase enzyme that hydrolyzes the substrate and generates a chemiluminescent signal proportional to the level of cAMP in the cell.

2 Materials Provided

List of Components	95-0119Y2-00103	95-0119Y2-00104
cAMP Hunter CHO-K1 FSHR Bioassay Cells (1.2 x10^6 cells/0.1 mL per vial)	2 vials	10 vials
cAMP Detection Kit for Bioassays		
cAMP Standard (250 µM) (mL)	0.2	1
cAMP Antibody Reagent (mL)	5	25
cAMP Lysis Buffer (mL)	7.6	38
Substrate Reagent 1 (mL)	2	10
Substrate Reagent 2 (mL)	0.4	2
cAMP Solution D (mL)	10	50
cAMP Solution A (mL)	16	80
AssayComplete™ cAMP Cell Assay Buffer	2 X 50 mL	4 X 50 mL
AssayComplete Cell Plating Reagent 2	1 X 100 mL	2 X 100 mL
Ultrapure IBMX	1 vial	1 vial
*Control Agonist: Follicle Stimulating Hormone (FSH)	1 vial	1 vial
96-well White, Clear Flat-Bottom, TC-Treated, Sterile Plates with Lid	2 plates	10 plates

Note: *Positive control used for QC testing of the bioassay cells in this kit as reflected in Certificate of Analysis.

3 Storage Conditions

cAMP Hunter CHO-K1 FSHR Bioassay Cells

Bioassay cells are shipped on dry ice and should arrive in a frozen state. To ensure maximum cell viability, upon receipt, the vials of bioassay cells must be stored in the vapor phase of liquid nitrogen. Please contact Technical Support (DiscoverXSupport@discovery.eurofinsus.com) immediately, if the cells received were already thawed.

If immediate storage in liquid nitrogen is not feasible, store vials at -80°C immediately upon arrival, but no longer than 24 hours.

Safety Warning: A face shield, gloves, and a lab coat should be worn at all times when handling frozen vials. The manufacturer of the cryovial recommends storing the vials in the vapor phase above the liquid nitrogen. Upon thawing, if liquid nitrogen is present in the cryovial, it rapidly converts back to its gas phase which can result in the explosion of the vial upon its removal from the liquid nitrogen tank.

AssayComplete Cell Plating Reagent 2 (CP2)

Upon receipt, store at -20°C. Once thawed, CP2 reagent can be stored at 4°C for up to 4 weeks. For longer-term storage (up to the expiration date listed on the kit certificate of analysis), the reagent should be aliquoted and stored at -20°C until needed. Once aliquoted, do not freeze-thaw the reagent more than two times. To make aliquots suitable for testing one assay plate each, 20 mL of reagent per aliquot can be dispensed and frozen down.

Control Agonist: Follicle Stimulating Hormone (FSH)

Store at -20°C until ready to use (up to the expiration date listed on the kit certificate of analysis). Centrifuge the vial prior to opening to maximize recovery and reconstitute as noted in the ligand datasheet with 0.1 mL of sterile deionized water to make a stock concentration of 100 ug/mL. Reconstituted ligand is stable for 1 week at 2-8°C or 12 months at -20 to -80°C, when reconstituted in buffer containing 0.1% BSA.

cAMP Detection Kit & cAMP Cell Assay Buffer

Upon receipt, store reagents at -20°C. We recommend that the Detection Reagent kit should be thawed from -20°C to room temperature for at least 24 hours prior to use. After thawing the kit to room temperature, leave it at 2-8°C overnight before use (total ~36 hours of thawing time before using the kit). Ensure that the reagents are equilibrated to room temperature before use in the assay for best performance.

After thawing, store reagents for up to 4 weeks at 2-8°C. For longer storage (up to the expiration date listed on the kit certificate of analysis), the reagent should be aliquoted and stored at -20°C in opaque containers until needed. Once aliquoted do not freeze-thaw more than twice.

Ultrapure IBMX

Store at -20°C until ready to use (up to the expiration date listed on the kit certificate of analysis). Centrifuge the vial prior to opening to maximize recovery and reconstitute as noted in the IBMX datasheet or as described in Step 6.2. Reconstituted IBMX is stable up to the expiration date listed, when stored at -20°C to -80°C. Avoid multiple freeze-thaw cycles.

96-Well Tissue Culture-Treated Assay Plates

Store at room temperature (23 - 25°C).

4 Additional Materials Required

Material	Ordering Information
96-well, V-Bottom, Untreated, Non-Sterile Dilution Plates	DiscoverX, Cat No. 92-0011, or similar
Multimode or luminescence plate reader	Refer to Instrument Compatibility Chart at discoverx.com/ instrument-compatibility
Single and multichannel micro-pipettors and pipette tips	
Sterile disposable reagent reservoir	ThermoFisher Scientific, Cat. No. 8094 or similar
50 mL and 15 mL Polypropylene tubes, sterile	

5 Protocol Schematic

In a 96-well tissue culture treated plate provided in the kit, perform the following steps:

Thaw cells using 1 mL of pre-warmed CP2

↓

Resuspend in a total of 12 mL of CP2

↓

Seed 100 µL of cells per well into a 96-well plate

↓

Incubate at 37˚C and 5% CO_2 for 48 hours

↓

Remove cell plating media from assay plate

↓

Immediately add 30 µL of pre-warmed cAMP Cell Assay Buffer

↓

Add 15 µL of the Follitropin alfa dilution series prepared in cAMP Cell Assay Buffer + IBMX

↓

Incubate at 37˚C and 5% CO_2 for 30 minutes

↓

Add 15 µL of cAMP Antibody Reagent

↓

Add 60 µL of cAMP Working Detection Solution

↓

Incubate for 1 hour at room temperature in the dark

↓

Add 60 µL of cAMP Solution A

↓

Incubate for 3 hours at room temperature in the dark

↓

Read chemiluminescent signal

'Created with BioRender.com'

Room temperature refers to a range of 23 - 25°C

6 Detailed Protocol

This assay is performed under aseptic conditions. Preparation of cells and Reference Standards/Test Samples are performed in a Biological Safety Cabinet following good aseptic technique. All appropriate materials are either certified as sterile or prepared aseptically. Prepared volumes may be scaled up or down if required.

6.1 Day 1: cAMP Hunter Bioassay Cells Preparation

The following protocol is for thawing and plating frozen cAMP Hunter CHO-K1 FSHR Bioassay cells from cryovials.

1) Before thawing the cells, ensure that all the necessary materials required are set up in the tissue culture hood. This includes:
    a) One 25 mL reagent reservoir
    b) One 15 mL conical tube
    c) A micropipettor (P1000) set to dispense 1 mL
    d) A multichannel pipette and tips set to dispense 100 µL
    e) An aliquot of AssayComplete™ Cell Plating 2 Reagent (CP2), **pre-warmed in a 37°C water bath for 15 minutes and equilibrated to room temperature**
    f) A 96-Well White, Clear bottom, Tissue Culture-Treated Sterile Assay Plate (provided with the kit)
2) Dispense 12 mL of CP2 into the 15 mL conical tube.
3) Remove the cryovial from liquid nitrogen and immediately place in dry ice.

DO NOT use heated water bath to thaw the vial. Wipe down the cryovial quickly with 70% ethanol and bring it into the tissue culture hood right away. DO NOT touch the sides or bottom of the vial to avoid thawing of the cell pellet through transfer of body heat.

4) Thaw the pellet by immediately adding 1 mL (using P1000) of pre-warmed CP2 from the 15 mL conical tube to the cryovial, thawing the cell pellet.
5) Add the medium slowly along the side of the wall of cryovial tube. Mix the cells gently by pipetting up and down several times to break up any clumps.
6) Transfer the cell suspension to the conical tube containing the remaining 11 mL of CP2. Remove any medium/suspension left in the tube to ensure complete recovery of all the cells from the vial.
7) Replace the cap on the conical tube and mix by gentle inversion 2-3 times to ensure that the cells are properly resuspended in the reagent, without creating any froth in the suspension. Immediately transfer the suspension into the sterile 25 mL reagent reservoir.
8) Using a manual 12-channel multichannel pipet, transfer 100 µL of the cell suspension to each well of the 96-well assay plate one row at a time, using reverse pipetting.
9) Mix cells in reagent reservoir by pipetting up and down 2-3 times gently before aspirating and dispensing cells into each subsequent row in the assay plate.
10) Replace the lid and leave the plate at room temperature for 15 minutes, to allow the cells to settle uniformly in the well and minimize potential edge effects.

11) Incubate the assay plate in humidified tissue culture incubator at 37°C and 5% CO2 for 48 ± 2 hours before proceeding with the assay.

6.2 Day 3: Sample Preparation

The following protocol is designed for testing purified biologics. The cAMP Hunter assays can also be run in the presence of high levels of serum or plasma without significantly impacting assay performance. Therefore, standard curves of control can typically be prepared in neat serum or plasma and added directly to cells without further dilution. For the best results, the optimized minimum required dilution (MRD) of crude samples should be empirically determined.

1) Reconstitute IBMX supplied in kit to a concentration of 500 mM by adding 0.5mL DMSO to supplied vial. To avoid condensation, centrifuge vial prior to opening. Mix well and generate small aliquots of stock solution. Store unused aliquots at –20°C. Do not freeze-thaw more than once.

2) Prepare 5 mL of ligand diluent (AssayComplete cAMP Cell Assay Buffer + 1500 µM IBMX; note that final concentration of IBMX in the assay will be 500 µM) by adding 15 µL of 500 mM IBMX to 4.985 mL of cAMP Cell Assay Buffer. Use this ligand diluent for preparing dilutions of Follitropin alfa and test samples.

3) Prepare Follitropin alfa dilution series at 3X the desired final concentration (Figure 3). Follitropin alfa (Gonal-F) is supplied as a 600 IU/mL (300 IU in 0.5 mL) stock solution. Top dose in dilution series is 0.006 IU/µL or 6 IU/mL in the assay: 3X = 18 IU/mL.

4) Add 15 µL of Follitropin alfa from 600 IU/mL stock to 485 µL of ligand diluent prepared in step 1 above to generate an 18 IU/mL working stock; and mix well.

5) Add 200 µL of prepared ligand diluent to well A2, 150 µL of ligand diluent to wells A3-A8 and 180 µL of ligand diluent to wells A9 & A10 of the master dilution plate.

6) Transfer 200 µL of 18 IU/mL Follitropin alfa into well A1 of a master dilution plate.

7) Using a clean tip, transfer 40 µL of 18 IU/mL Follitropin alfa from well A1 into well A2 for a 6-fold dilution and mix by pipetting up and down several times.

    a) Replace the pipette tip and transfer 150 µL from well A2 into A3 for a 2-fold dilution and mix by pipetting up and down several times. Repeat 2-fold serial dilutions until well A8 is reached.

    b) Using a clean tip, transfer 20 µL from well A8 to well A9 for a 10-fold dilution and mix by pipetting up and down several times. No sample is transferred to A10 as this will serve as a negative control.

8) Prepare each Follitropin alfa test sample using the same dilution series as Follitropin alfa in a new row of the master dilution plate.

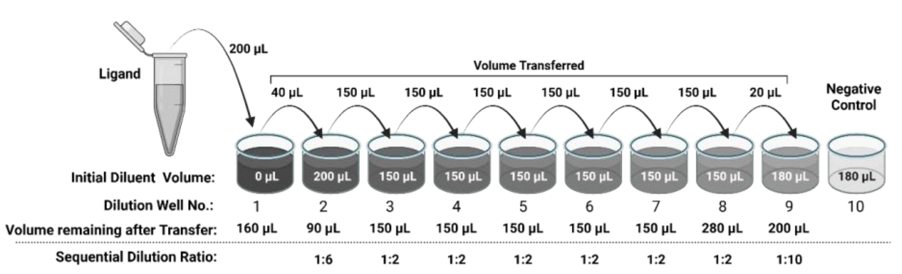


Figure 3: Preparation of Follitropin alfa Dilution Series

6.3 Day 3: Assay Plate Preparation

1) Remove assay plates with cells from incubator.
2) With a flicking motion, remove liquid from the assay wells in the plate. Then, place the assay plate upside-down on a lab-grade low lint tissue/paper towel. Spin the assay plate in a plate centrifuge set at 1000 rpm. Stop once it reaches 1000 rpm. Alternately, completely remove the Cell Plating Reagent from each assay plate by carefully aspirating the wells.
3) Immediately add 30 µL of pre-warmed AssayComplete cAMP Cell Assay Buffer to all empty wells of the plate.
4) Transfer 15 µL of the Follitropin alfa (Gonal-F) dilutions series (prepared in section Day 3: Sample Preparation) from the master dilution plate to the appropriate wells in the assay plate as indicated in the Representative Assay Plate Map (Figure 4: Representative Assay Plate Map).
5) Transfer 15 µL of test sample dilution series from dilution plate to the appropriate wells in the assay plate as indicated in the Representative Assay Plate Map (refer to Figure 4: Representative Assay Plate Map; Test Sample wells).
6) Incubate assay plate at 37°C and 5% CO_2 incubator for 30 minutes.

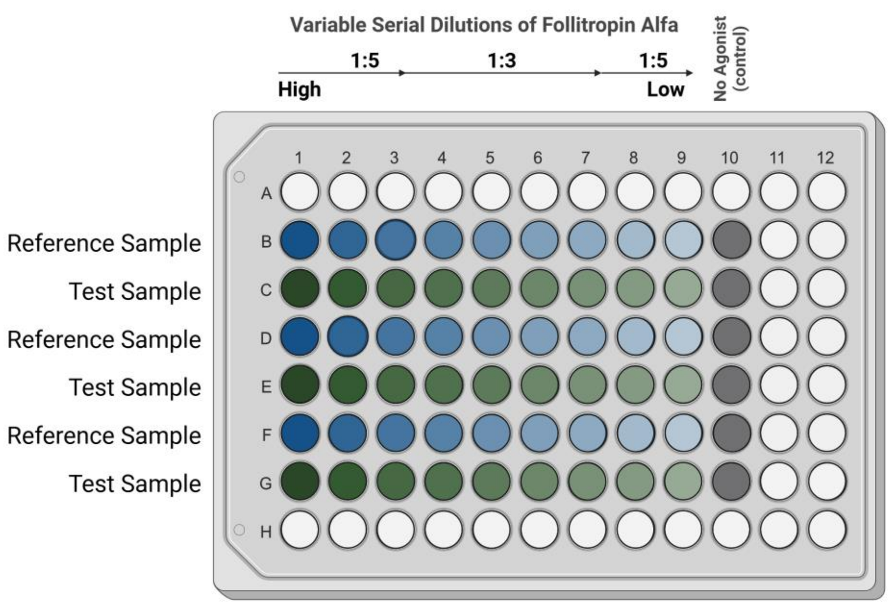


Figure 4: Representative Assay Plate Map

This plate map shows a 9-point dilution curve with three data points at each concentration for one reference and one test sample per plate, with a variable serial dilution scheme.

6.4 Day 3: Preparation of FSH Control Ligand

Use the FSH Control ligand to verify your bioassay kit's performance before running the full potency assay.

Follow the same general procedure as the potency assay in sections 0 and 6.3 but replace the Follitropin alfa preparation (section Day 3: Sample Preparation) with the preparation of the FSH control described in this section. You only need to run a single curve for the FSH control.

The **cAMP standard** can also be used to confirm that the assay is performing correctly.

1) Prepare ligand diluent (cAMP Cell Assay buffer + 1500 µM IBMX) by adding 15 µL of 500 mM IBMX to 4.985 mL of cAMP Cell Assay Buffer. Note that final concentration of IBMX in the assay will be 500 µM.
2) Reconstitute FSH: Add 100 µL of sterile deionized water to 10 µg of lyophilized powder of FSH protein to make a stock of 100 µg/mL.
3) Prepare intermediate and working stock solutions of FSH as shown in the table below. Note that the working dilution is prepared at 3X the desired top concentration of FSH in the ligand dilution series.

FSH Sample	Concentration	Ligand vol (µL)	Diluent vol (µL)
Intermediate Dilution 1	30 µg/mL	12 µL of 100 µg/mL stock	28
Working stock	3 µg/mL	15 µL of 30 µg/mL	135

4) Prepare FSH serial dilutions (3-fold serial dilutions) as a 3X stock:
   a) Add 100 µL of ligand diluent (from Step 1)) to Wells A2-A12 of the master dilution plate.
   b) Transfer 150 µL of FSH working stock to Well A1 of the master dilution plate.
   c) Transfer 50 µL of FSH working stock from Well A1 into Well A2 and mix well by pipetting up and down several times. Change tips and transfer 50 µL from Well A2 into Well A3 and mix well. Continue this dilution series until Well A11 is reached. No ligand is added to Well A12 as it will serve as the negative control.

6.5 Day 3: cAMP Detection

During the 30-minute agonist incubation period, prepare cAMP working detection solution in a separate 15 mL polypropylene tube by mixing 19-parts of cAMP Lysis Buffer, 5-parts of Substrate Reagent 1, 1-part Substrate reagent 2, and 25-parts of cAMP Solution D.

Refer to Table below for the volume of each component required for one 96-well plate. Adjust volumes according to your assay set-up, keeping the ratios consistent. Store in the dark before use.

Components	Volume Ratio	Volume per 96-Well Plate (mL)
cAMP Lysis Buffer	19.0	3.8
Substrate Reagent 1	5.0	1.0
Substrate Reagent 2	1.0	0.2
cAMP Solution D	25.0	5.0
Total Volume		**10**

cAMP Working Detection Solution is light sensitive, thus storage and incubation in the dark is necessary.

Following agonist incubation, add 15 µL of the cAMP Antibody Reagent to all wells of the assay plate

1. Add 60 µL of cAMP Working Detection Solution to all wells of the assay plate. Do not pipette up and down in the wells to mix or vortex plates.
2. Incubate assay plate for 1 hour at room temperature in the dark.

Add 60 µL of cAMP Solution A to all wells of the assay plate. Do not pipette up and down in the wells to mix or vortex plates.

3. Incubate assay plate for 3 hours at room temperature in the dark.

cAMP Working Detection Solution is light sensitive, thus storage and incubation in the dark is necessary.

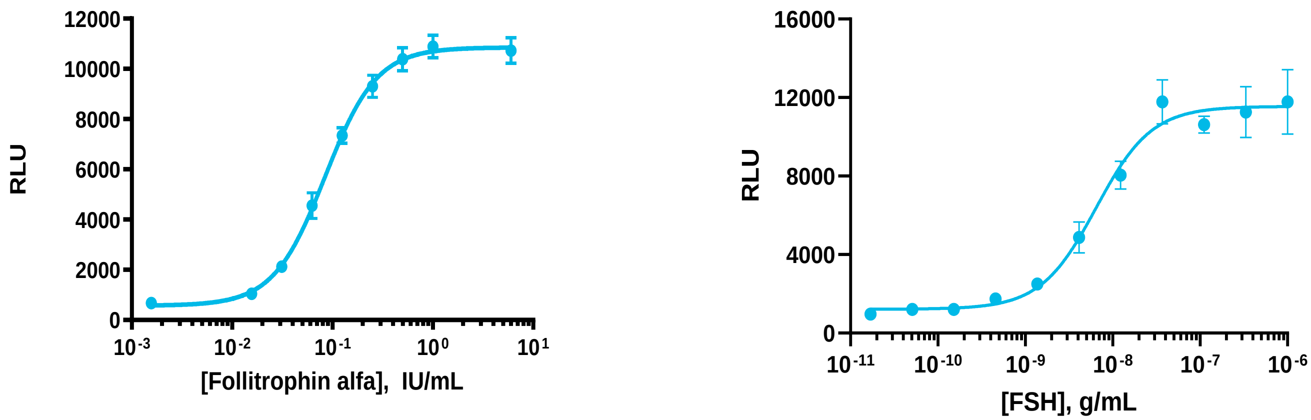
4. Read sample on a standard luminescence plate reader at 0.1 to 1 second/well for photomultiplier tube (PMT) readers or 5-10 seconds for imager.

7 Typical Results

The following graph is an example of a typical dose-response curve for the Follitropin alfa Bioassay generated using the protocol outlined in this manual. The data shows a potent, dose-dependent increase in cAMP production when treated with (A) Follitropin alfa and (B) FSH. The plate was read on the EnVision® Multimode Plate Reader, with a 0.2 sec/well integration time, and data analysis was conducted using GraphPad Prism.

A

B



Follitropin alfa	
Signal to Background (S/B)	EC_{50} (IU/mL)
18.3	0.042

FSH	
Signal to Background (S/B)	EC_{50} (ng/mL)
12.3	6.5

Figure 5: Typical results for Follitropin alfa and FSH

Representative dose-response curves and EC_{50} and assay window (Signal/Background, S/B) for (A) Follitropin alfa- and (B) FSH- mediated FSHR activation, as measured in this bioassay.

8 Troubleshooting Guide

Problem	Cause	Solution
No response	Improper thawing procedure	Refer to thawing instructions in this user manual. Thawing process can have a significant effect on cell viability.
	Improper ligand used or improper ligand incubation time	See certificate of analysis for recommended ligand and assay conditions.
	Improper preparation of ligand (agonist or antagonist)	Refer to specific datasheet to ensure proper handling, dilution and storage of ligand.
	Improper time course for induction	Optimize time course of induction with agonist and antagonist.
Decreased response	Cells are not adherent and exhibit incorrect morphology	Confirm adherence of cells using microscope.
Low or no signal	Improper preparation of detection reagents	Detection reagents should ideally be prepared just prior to use and are sensitive to light.
	Problem with microplate reader	Microplate reader should be in luminescence mode. Read at 0.1-1 seconds/well.
Experimental S:B does not match datasheet value	Incorrect incubation temperature	Confirm assay conditions. Check and repeat assay at correct incubation temperature as indicated on the assay datasheet.
	Improper preparation of ligand (agonist or antagonist)	Some ligands are difficult to handle. Confirm the final concentration of ligands.
EC_{50} is right-shifted	Improper ligand handling or storage	Make sure ligands are stored and incubated at the proper temperature.
	Difference in agonist binding affinity	Confirm that the ligand used is comparable to the ligand in the certificate of analysis.
	Problems with dynamic range or dilutions	Changing tips during serial dilutions can help to avoid carryover.
		Include a cAMP standard curve in the assay to ensure that the ligand dose curve is in the dynamic range of the cAMP detection kit. Moving out of the dynamic range may shift the EC_{50} of ligands.
High variability between replicates	Instrument calibration	Ensure dispensing equipment is properly calibrated, and proper pipetting technique is used.

For questions on using this product, please contact Technical Support at discoverx.com/support/

9 Document Revision History

Revision Number	Date Released	Revision Details
0	April 2017	New Document
1	September 2025	• Updated cell assay buffer name to AssayComplete cAMP Cell Assay Buffer • Updated drug name from Gonal-F to Follitropin alfa • Added protocol for preparing positive control (FSH) • Corrected the timelines for the assay protocol

10 Limited Use License Agreement

These products may be covered by issued US and/or foreign patents, patent application and subject to Limited Use Label License.

Please visit discoverx.com/license for limited use label license and trademark information for this product.

cAMP Hunter™ Follitropin alfa Bioassay Kit User Manual 70-392 Rev 1